

Fan Vote Uncovering, Rule Evaluation, and Fairness-Centered Optimization

Summary

Aiming at the core issues in *Dancing with the Stars* (DWTS)—undisclosed fan votes, rule-induced unfairness, divergent judge–fan preferences, and frequent controversies—this study develops a comprehensive quantitative framework of “vote inversion–rule comparison–characteristic analysis–mechanism optimization” based on inverse problem solving, **Monte Carlo** simulation, **mixed-effects** modeling, and mechanism design, with robustness verified **via sensitivity analysis**.

For **Problem 1**, unobservable fan vote shares are treated as **latent variables** in the simplex space. A differentiated inversion model is constructed: quadratic programming with **slack variables** for the percent-based system (S3–S27), and **Dirichlet-guided** sampling with importance weighting for the rank-based system (S1–S2, S28–S34). Uncertainty is quantified using confidence intervals and **effective sample size (ESS)**, with the model achieving a 100% hit rate in reproducing eliminations.

For **Problem 2**, fan vote shares estimated in Problem 1 are modeled as **Beta-distributed random variables**. **Monte Carlo simulation** shows the percent-based rule favors fan votes, while the rank-based rule emphasizes judges. The Judges’ Second-Review Mechanism moderates extreme outcomes, reducing survival probabilities of controversial contestants and **alleviating “technical-popularity conflict.”**

For **Problem 3**, **mixed-effects models** reveal key divergences: professional dancers significantly influence judges’ scores but not fan votes; age positively affects fans yet negatively impacts judges; industry differences drive preference gaps, and fan support is independent of technical performance.

For **Problem 4**, the New rule is proposed, integrating STF, z-score standardized composite scoring, and **“Strike & Cap”** disciplinary constraints. Simulation results show New rule reduces controversy weeks to 10.0%, **corrects unfair survivals**, and maintains over 72% consistency with original rules.

Sensitivity analysis confirms robustness: perturbing key hyperparameters like constraint margin Δ , concentration coefficient γ , and prior weight w_{prev} leaves **core metrics** unchanged. This study provides actionable insights for reality TV scoring mechanism design.

Keywords: Beta distribution; Mixed-Effects Model; Monte Carlo Simulation; feature analysis; maximum entropy

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1 Introduction

1.1 Background

Tennis performance varies within a match due to physical, tactical, and psychological factors, leading observers to describe play in terms of “momentum” or “flow.” With point-by-point data, we model match dynamics and compare observed patterns with stochastic benchmarks to determine whether apparent momentum reflects real shifts in competitiveness or random variation.

1.2 The Description of the Problem

To address the core challenges of fan vote estimation, rule comparison, factor analysis, and mechanism optimization in Dancing with the Stars (DWTS), our specific objectives are as follows:

Task 1: Develop a mathematical model to estimate weekly fan vote shares for each contestant, verify consistency with observed eliminations, and quantify the certainty of estimates.

Task 2: Compare rank-based and percent-based vote-combining methods across seasons, analyze their bias toward fan votes, evaluate impacts on controversial contestants, and recommend an optimal method (including the "Bottom 2 + Judges' choice" mechanism).

Task 3: Construct a model to quantify the impact of professional dancers and celebrity characteristics (age, industry, etc.) on competition outcomes, and examine if their effects differ between judges' scores and fan votes.

Task 4: Propose a more fair/exciting weekly scoring and elimination system integrating fan votes and judges' scores, with supporting evidence for producers.

1.3 Our work

Our solution mindmap is shown below.

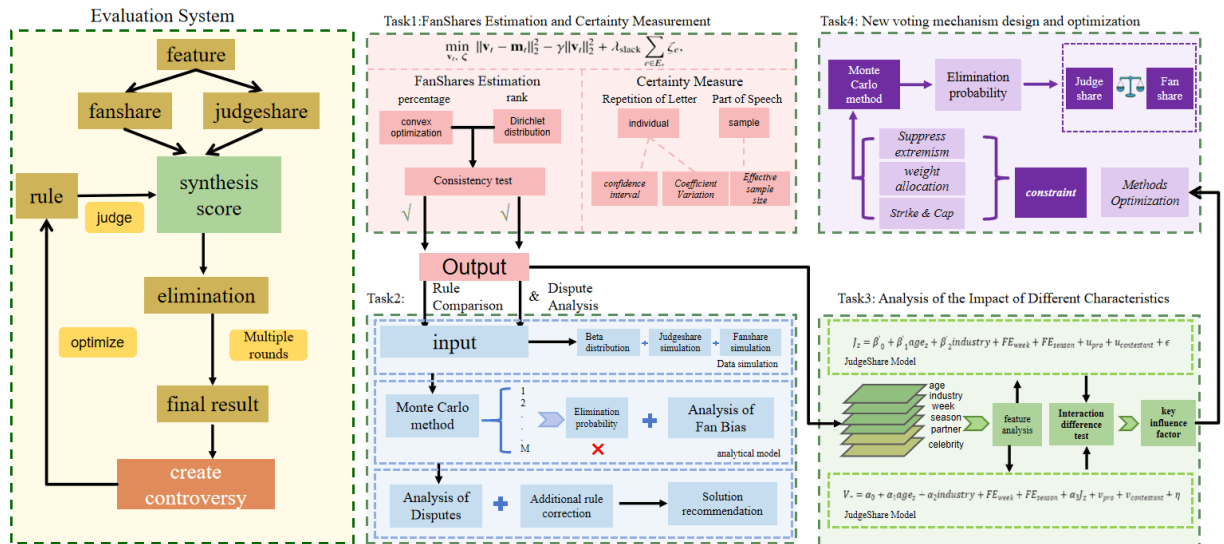


Figure 1: Our work

2 Assumptions

To simplify the problem, we make the following assumptions, each of which is reasonably justified.

Assumption 1: Fan vote shares of contestants in different weeks are relatively independent, with no direct causal correlation.

Justification: Short-term fluctuations in fan preference are absorbed into the uncertainty of vote-share estimation. Modeling explicit sequential dependencies would require external information that is unavailable in the dataset. Hence, we assume independence to balance feasibility and rationality.

Assumption 2: Judges' scores objectively reflect contestants' dance technique and on-stage performance, with no systematic bias unrelated to performance.

Justification: Subjective factors are unmeasurable given the existing data. Assuming score objectivity allows judges' scores to serve as a valid proxy for "professional evaluation" in the model.

Assumption 3: The official aggregation rules are implemented strictly, without unrecorded arbitrary adjustments.

Justification: Unrecorded rule changes would introduce unquantifiable bias into the analysis. Assuming strict rule execution ensures that the model's rule-based calculations align with the show's actual operation.

3 Notations

Table 1: Notations Table

Symbol	Definition
A_t	Set of contestants still competing in week t
$p_{i,t}$	Judges' score share of contestant i in week t
$r_{i,t}$	Fan-vote rank of contestant i in week t
$R_e^{(k)}$	Rank of the eliminated contestant under the k -th sample
T	Rule-implied elimination threshold (rank cutoff)
$X_{s,w,i}$	Sampled (pre-normalization) fan vote value for contestant i in season s , week w
$\tilde{V}_{s,w,i}$	Normalized fan vote share for contestant i in season s , week w
$\tilde{V}_{s,w,i}^{(m)}$	Vote-share realization for contestant i in Monte Carlo trial m
$\hat{p}_{\text{elim}}(i; s, w, r)$	Estimated elimination probability of contestant i under rule r in season s , week w
FE_{week}	Week fixed effects
$\text{FE}_{\text{season}}$	Season fixed effects
u_{pro}	Pro-level random effect
$u_{\text{contestant}}$	Contestant-level random effect
$\mathbb{I}[\text{source} = \text{Judge}]$	Channel indicator: 1 for judges observations, 0 for fan-vote observations

4 Modeling and Solving of Problem 1

4.1 Problem Analysis

Each week of *Dancing with the Stars* provides the active couples, judges' scores, and the elimination outcome^[1], while fan voting is unobserved. Thus, Problem 1 is an inverse problem: infer weekly fan support consistent with the published results.

Let A_t be the active set in week t , and let $\mathbf{v}_t = \{v_{i,t}\}_{i \in A_t}$ denote the latent fan vote-share vector on the simplex. We validate $\mathbf{v}_t$ by re-applying the season's aggregation rule (percent- or rank-based) and checking whether the observed eliminated couple(s) fall within the model-implied high-risk set. Since eliminations provide only partial information, we additionally quantify uncertainty to reflect how tightly the weekly constraints identify $\mathbf{v}_t$. Normalized vote shares are sufficient because the rules depend on relative support rather than absolute vote totals.

To clarify the overall reconstruction pipeline before presenting detailed formulations, Figure X summarizes the sequential vote-share estimation and elimination-consistency validation procedure.

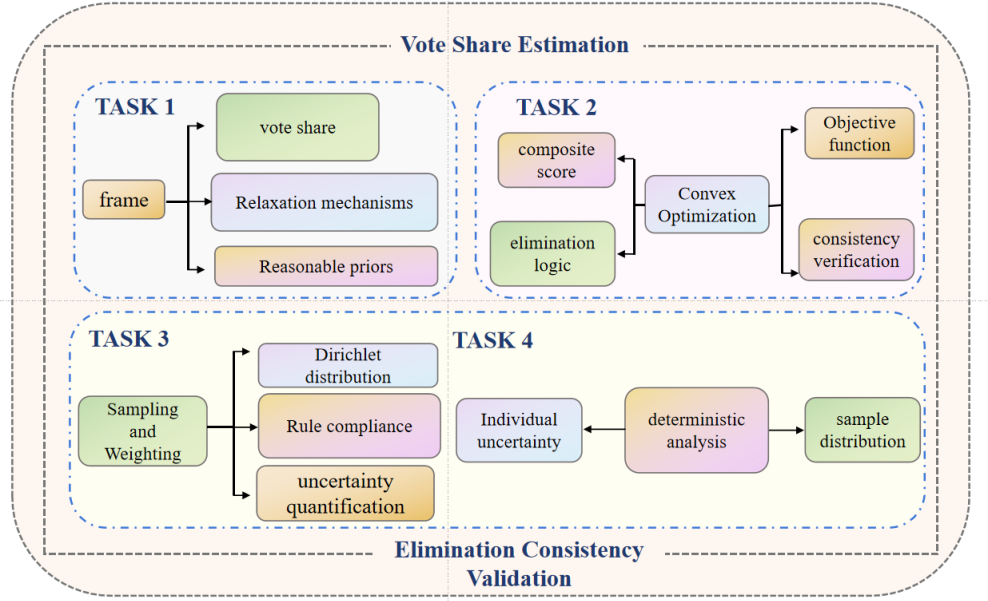


Figure 2: Vote-share estimation pipeline: inference, uncertainty, and consistency validation.

4.2 Model Preparation: Data Handling and Latent Variables

Active sets and weekly records. The dataset records judges' scores as 0 after a couple is eliminated. Those zeros do not represent real judging in subsequent weeks and would inject pseudo-information if retained. All post-elimination records are therefore removed, and each week is analyzed only on the effective active set A_t . Entries marked N/A are treated as missing. The eliminated set in week t is denoted by $E_t \subseteq A_t$, and the survivors are $R_t = A_t \setminus E_t$.

Weekly fan vote shares. Fan support in week t is represented by a vote-share vector on the simplex:

$$v_{i,t} \geq 0, \quad \sum_{i \in A_t} v_{i,t} = 1. \quad (1)$$

This normalization absorbs the unknown total number of votes cast in week t , while preserving the relative magnitudes that enter the official aggregation rules.

Mapping information across weeks. Because eliminations change the active set, the dimension of $\mathbf{v}_t$ varies over time. When information from week $t - 1$ is carried forward, it is restricted to the current active set and renormalized:

$$\tilde{v}_{i,t-1} = \frac{\hat{v}_{i,t-1}}{\sum_{j \in A_t} \hat{v}_{j,t-1}}, \quad i \in A_t. \quad (2)$$

For $t = 1$, the initialization uses a flat prior on the opening cast: $\tilde{\mathbf{v}}_0 = \mathbf{u}_1$ with $u_{i,1} = 1/|A_1|$.

Season-level baseline (persistent fanbase). Weekly eliminations typically provide weak identification of $\mathbf{v}_t$. A purely week-to-week smoothness prior tends to keep solutions close to uniform when constraints are loose. To represent persistent popularity within a season, a season baseline $\mathbf{b}_t$ is maintained by an exponential moving average (EMA) update:

$$\mathbf{b}_t \leftarrow (1 - \rho) \mathbf{b}_{t-1} + \rho \hat{\mathbf{v}}_t, \quad (3)$$

with $\rho \in (0, 1)$. When used as prior information in a later week, $\mathbf{b}_{t-1}$ is mapped to the current active set using the same restriction-and-renormalization operation as in (2). This baseline prevents the reconstruction from repeatedly collapsing toward a near-uniform share in weeks where the elimination constraints are permissive.

4.3 Model Construction

4.3.1 System I: Percent-Based Aggregation (Seasons 3–27)

In percent-based seasons, the show combines the share of judges' points and the share of fan votes. Let $J_{i,t}$ be the total judges score for couple i in week t , and define the judges share

$$p_{i,t} = \frac{J_{i,t}}{\sum_{j \in A_t} J_{j,t}}, \quad i \in A_t. \quad (4)$$

The combined score is

$$S_{i,t} = p_{i,t} + v_{i,t}. \quad (5)$$

Elimination constraints with margin and slack. Under an idealized interpretation of the rule, eliminated couples should not exceed survivors in combined score. A small separation margin $\Delta > 0$ is imposed for numerical stability, and nonnegative slack variables $\zeta_e \geq 0$ are introduced to preserve feasibility in special-rule weeks:

$$(p_{s,t} + v_{s,t}) - (p_{e,t} + v_{e,t}) \geq \Delta - \zeta_e, \quad \forall e \in E_t, \forall s \in R_t. \quad (6)$$

When $\zeta_e = 0$, the observed elimination is consistent with the strict inequality system; when $\zeta_e > 0$, the week requires an exception relative to the simplified constraint set.

Regularized reconstruction objective. Weekly constraints alone generally do not identify $\mathbf{v}_t$ uniquely. The reconstruction is therefore regularized by a prior target $\mathbf{m}_t$ combining three components on the current active set: (i) the mapped previous-week estimate $\tilde{\mathbf{v}}_{t-1}$, (ii) the mapped season baseline $\tilde{\mathbf{b}}_{t-1}$, and (iii) a weak uniform anchor $\mathbf{u}_t$:

$$\mathbf{m}_t = w_{\text{prev}} \tilde{\mathbf{v}}_{t-1} + w_{\text{base}} \tilde{\mathbf{b}}_{t-1} + w_{\text{unif}} \mathbf{u}_t, \quad w_{\text{prev}} + w_{\text{base}} + w_{\text{unif}} = 1. \quad (7)$$

In addition, mild *concentration encouragement* is used to avoid unrealistically uniform shares in weakly constrained weeks. On the simplex, a larger $\|\mathbf{v}_t\|_2^2$ corresponds to a more peaked distribution, so the objective includes a negative quadratic term $-\gamma\|\mathbf{v}_t\|_2^2$ with $\gamma \in [0, 1)$ to preserve convexity. The week- t estimate is obtained by

$$\min_{\mathbf{v}_t, \zeta} \|\mathbf{v}_t - \mathbf{m}_t\|_2^2 - \gamma\|\mathbf{v}_t\|_2^2 + \lambda_{\text{slack}} \sum_{e \in E_t} \zeta_e, \quad (8)$$

subject to

$$v_{i,t} \geq 0, \quad \sum_{i \in A_t} v_{i,t} = 1, \quad \zeta_e \geq 0, \quad \text{and} \quad (6). \quad (9)$$

This remains a quadratic program with linear constraints.

Week-level margin diagnostic. For interpretability, the separation implied by $\hat{\mathbf{v}}_t$ is summarized by the observed margin between the weakest survivor and the strongest eliminated couple:

$$\text{gap}_t = \min_{s \in R_t} S_{s,t} - \max_{e \in E_t} S_{e,t}. \quad (10)$$

When the model is consistent without slack, gap_t is expected to be near (and above) Δ .

4.3.2 System II: Rank-Based Aggregation (Seasons 1–2 and 28–34)

In rank-based seasons, judges and fans each produce a rank and the ranks are added. Let $j_{i,t}$ be the judges rank (1 = best judges score) and $r_{i,t}$ the fan rank (1 = most votes). The combined rank is

$$R_{i,t} = j_{i,t} + r_{i,t}, \quad (11)$$

where larger $R_{i,t}$ indicates worse standing.

Era-specific Bottom- T criterion. **Definition (first use):** throughout the paper, *Bottom- k* denotes the set of the k worst couples under the model-implied weekly ordering (Percent: smallest $S_{i,t}$; Rank: largest $R_{i,t}$). The elimination mechanism differs across rank-based eras:

Seasons 1–2: elimination aligns with the single worst combined rank (Bottom-1), so $T = 1$.

Seasons 28–34: the show forms a bottom-two set by combined ranks and then eliminates within that set, so $T = 2$.

Sampling-based reconstruction. Because share-to-rank mapping is non-smooth, we reconstruct vote shares via guided simplex sampling with importance weighting. We draw candidates from a Dirichlet proposal centered at a mixed prior (previous estimate and season baseline), convert each draw to fan ranks, and define rule deviation by whether the observed elimination lies outside the bottom- T set. Samples are reweighted for rule consistency and temporal plausibility, with a mild non-uniformity preference via $\|\mathbf{v}\|_2^2$.

4.4 Model Solution

4.4.1 Sequential inference framework

We process each season chronologically. At week t , we form the active set A_t after removing post-elimination records, compute judges' shares $\mathbf{p}_t$ over A_t , and read the eliminated set E_t . We then estimate $\hat{\mathbf{v}}_t$, pass it forward as the prior for week $t + 1$, and update the season baseline $\mathbf{b}_t$ via (3).

4.4.2 Percent era (S3–S27) and Rank era (S1–S2 and S28–S34)

Percent era (S3–S27): For percent-based seasons, week t inference solves the quadratic program (8)–(9). When $E_t = \emptyset$, elimination constraints are absent and $\hat{\mathbf{v}}_t$ is determined by the mixture prior (7) and the concentration term. When $|E_t| > 1$, constraints (6) are applied to each eliminated couple.

Rank era (S1–S2 and S28–S34): For week t , we draw K Dirichlet samples^[2] on the simplex, convert them to fan ranks, form combined ranks, and check whether eliminated couples fall in the era-specific bottom- T set. Samples are importance-weighted by rule consistency, proximity to the mixed prior, and mild concentration preference, and $\hat{\mathbf{v}}_t$ is the weighted mean.

4.4.3 Consistency Metrics

Consistency is evaluated in-sample by comparing observed eliminations with the model-implied risk set:

Bottom-2 hit rate: the fraction of elimination weeks in which at least one eliminated couple falls within the bottom two under the model-implied ordering;

Bottom-1 accuracy: the fraction of elimination weeks in which the eliminated couple is the single worst under the model-implied ordering.

Bottom-2 is the primary cross-era criterion because Seasons 28–34 restrict elimination to a bottom-two set.

Figure 3 reports era-level consistency with 95% confidence intervals: Bottom-2 hit rate is essentially 1.0 in all eras, while Bottom-1 accuracy is slightly lower under Rank+Save, consistent with judges selecting within Bottom-2. The wider CI for Rank (S1–S2) reflects its smaller sample size ($N = 11$) versus the Percent era ($N = 194$).

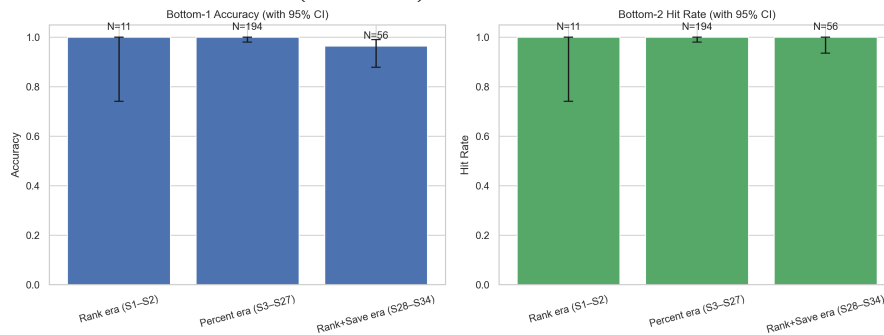


Figure 3: Left: Bottom-1 accuracy; Right: Bottom-2 hit rate

4.4.4 Uncertainty Quantification

Contestant-week uncertainty (CI and CV). For each couple-week, we report a 95% interval and width, plus $CV = \text{std}/\text{mean}^{[3]}$. Rank-era intervals use weighted sample percentiles. Percent-era uncertainty is obtained by constraint-aware sampling around $\hat{\mathbf{v}}_t$ on the simplex, retaining draws that satisfy (6) (within tolerance).

Uncertainty scale summary (numbers). Table 2 reports median and IQR uncertainty on elimination weeks, aggregated over contestant-weeks.

Table 2: Typical uncertainty scale on elimination weeks (median [IQR]).

Scheme	weeks	CI width	CV	ESS (week)	hard_pass_rate
Percent	194	0.095 [0.086, 0.104]	0.219 [0.176, 0.253]	6591 [4825, 8232]	0.220 [0.161, 0.274]
Rank	67	0.215 [0.187, 0.257]	0.583 [0.505, 0.654]	4968 [2318, 6239]	0.444 [0.213, 0.545]

Figure 4 visualizes Season 21 contestant-week uncertainty via CI-width heatmap. Uncertainty is heterogeneous within weeks and tends to increase late in the season for top contenders, consistent with weaker identifiability when many vote-share configurations remain feasible.

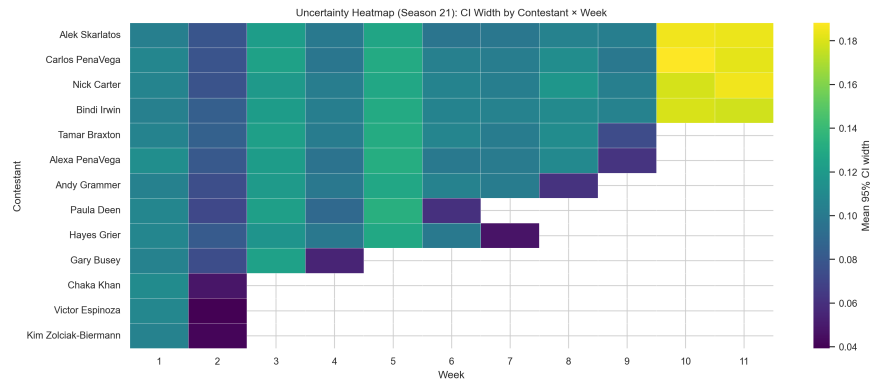


Figure 4: Uncertainty heatmap (Season 21): CI Width by Contestant $\times$ Week

Figure 5 plots Season 21 top-6 vote-share trajectories with 95% bands. The paths show week-to-week smoothing, while bands widen when outcomes weakly constrain feasible vote shares (close competitions), indicating multiple latent configurations can explain the same elimination.

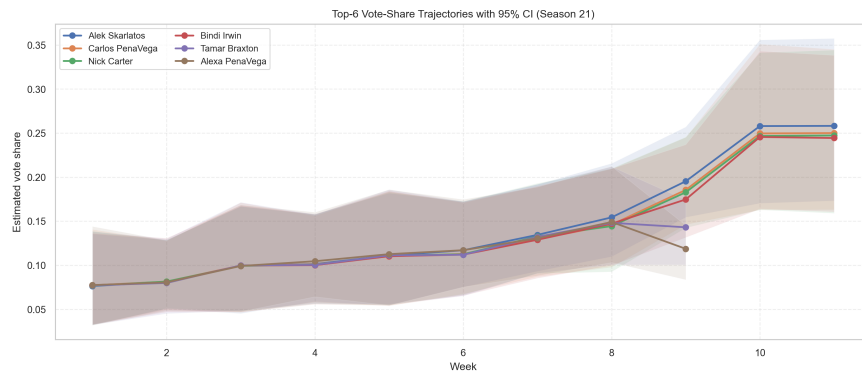


Figure 5: Top-6 Vote-Share Trajectories with 95% CI (Season 21)

Week-level diagnostics and interpretation. Two week-level quantities summarize how strongly the observed elimination restricts the vote shares:

ESS / accepted-sample size: the number of samples retained (or, in weighted sampling, the effective sample size $ESS = 1 / \sum_k w_k^2$). Larger values indicate more feasible solutions.

hard_pass_rate: the fraction of samples that satisfy the rule exactly (zero rule deviation).

Higher **hard_pass_rate** indicates a broader feasible region on the simplex; this often yields *larger* uncertainty because many vote-share vectors satisfy the elimination constraints. When **hard_pass_rate** is low, feasibility narrows and intervals shrink.

For the Percent era, Figure 6 links elimination “closeness” to identifiability. The left panel shows the gap distribution, with many weeks decided by small separations between the weakest survivor and strongest eliminated. The right panel highlights weeks with minimal slack/rule deviation, i.e., near-boundary (or multi-elimination) cases useful for stress-testing the reconstruction.

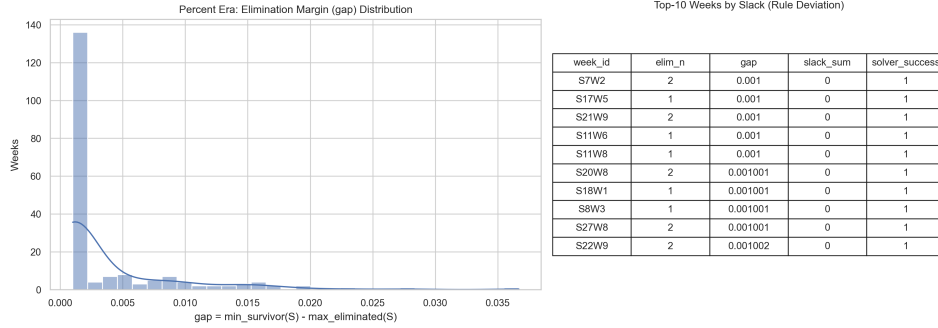


Figure 6: Left:Percent Elimination Margin (gap) Distribution Right: Top-10 Weeks by Slack

Table 3: Key hyperparameters used in Problem 1 (implementation settings)

Percent-QP / Prior	Value	Rank-Sampling	Value
Separation margin Δ	1×10^{-3}	Sample size K	12000
Slack penalty λ_{slack}	1×10^4	Prior distance weight β	12.0
Mixture weights ($w_{\text{prev}}, w_{\text{base}}, w_{\text{unif}}$)	(0.75, 0.23, 0.02)	Rule weight η_{rank}	80.0
Baseline EMA rate ρ	0.18	Dirichlet scale α_0	10.0
Concentration term γ	0.35	Concentration weight γ_{rank}	0.8

5 Modeling and Solving of Problem 2

5.1 Problem analysis

Problem 2 is a *system-level comparison and impact evaluation*. Building on Question 1, which provides the latent fan vote shares with uncertainty (mean and variability), we address:

- How different aggregation rules combining *judges' scores* + *fan votes* (Percent vs. Rank) change weekly eliminations and potentially the overall season outcome.

- Whether “controversial” contestants (high fan support but low judges’ evaluations, or vice versa) have systematically different fates under different rules.
- Whether an additional intervention mechanism—*Judges’ Save* within Bottom-2—can moderate controversial outcomes, and how to tune it for practical use.

Fan voting is hidden and Q1 provides distributions, so point estimates would make eliminations nearly deterministic and preclude risk assessment. We therefore propagate uncertainty and evaluate rules via Monte Carlo simulation.

5.2 Rules and simulation setup

5.2.1 Notation

Let $J_{s,w,i} \in [0, 1]$ be the normalized judges’ score of contestant i in season s , week w . Let $\tilde{V}_{s,w,i} \in [0, 1]$ be a sampled fan vote share in that week (renormalized so that $\sum_{i \in \mathcal{A}_{s,w}} \tilde{V}_{s,w,i} = 1$ over the active set $\mathcal{A}_{s,w}$). Let $E_{s,w}$ be the observed number of eliminations in week (s, w) .

5.2.2 Rule definitions

Percent rule (cardinal aggregation). We define the combined score

$$S_{s,w,i}^{(\text{pct})} = J_{s,w,i} + \tilde{V}_{s,w,i}, \quad (12)$$

and eliminate the $E_{s,w}$ contestants with the smallest $S^{(\text{pct})}$.

Rank rule (ordinal aggregation). Within each week, convert both components to ranks (1 = best):

$$R_{s,w,i}^{(J)} = \text{rank}(J_{s,w,i}), \quad R_{s,w,i}^{(V)} = \text{rank}(\tilde{V}_{s,w,i}), \quad (13)$$

then aggregate by

$$S_{s,w,i}^{(\text{rk})} = R_{s,w,i}^{(J)} + R_{s,w,i}^{(V)}, \quad (14)$$

and eliminate the $E_{s,w}$ contestants with the *largest* $S^{(\text{rk})}$ (worst total rank).

Judges’ Save (Percent + Bottom-2 + probabilistic judges’ choice). We implement Judges’ Save as a *controlled modification of the Percent ordering*:

1. Use $S^{(\text{pct})}$ to form a weekly ordering and identify the Bottom-2 contestants (the two lowest $S^{(\text{pct})}$).

2. Restrict the final decision to the Bottom-2. Let A, B be the Bottom-2 and let J_A, J_B be their judges’ scores. We model the probability that A is eliminated by^[4]

$$\mathbb{P}(A \text{ eliminated}) = \frac{1}{1 + \exp\{-\gamma(J_B - J_A)\}}, \quad (15)$$

where $\gamma > 0$ controls how strongly judges follow score differences. Larger γ means a more deterministic, score-driven save.

5.2.3 Monte Carlo evaluation (weekly risk and season survival)

For each elimination week (s, w) and each rule r , we run M Monte Carlo trials:

1. Sample fan vote shares for active contestants using the uncertainty output from Question 1, renormalize to obtain $\tilde{\mathbf{V}}_{s,w}$.
2. Apply rule r to determine the eliminated set.
3. Count how often each contestant is eliminated to estimate elimination risk $\hat{p}_{\text{elim}}(s, w, i | r)$.

Simulation-quality screening (pass rate). In forward simulation, some Monte Carlo paths may reach weeks where too few contestants remain to apply the observed $E_{s,w}$ eliminations (e.g., earlier eliminations under alternative rules). We define

$$\text{pass_rate}(s, w, r) = \frac{n_{\text{valid}}(s, w, r)}{M}, \quad (16)$$

where n_{valid} is the number of trials in which the week-step is executable under rule r . To avoid unstable estimates driven by very few valid trials, we restrict comparisons to elimination weeks with pass rate ≥ 0.2 , and we use *conditional* elimination probability:

$$\hat{p}_{\text{elim}}(s, w, i | r) = \frac{\#\{\text{valid trials eliminating } i\}}{n_{\text{valid}}(s, w, r)}.$$

We also compute each contestant's *observed-horizon survival probability* under each rule: the probability of remaining uneliminated when simulating forward over the observed season weeks and elimination schedule^[4].

5.3 Global comparison: Percent vs Rank (and Save)

5.3.1 Do Percent and Rank usually agree?

Let the *Top-1 risk candidate* in week (s, w) be the active contestant maximizing $\hat{p}_{\text{elim}}(s, w, i | r)$. Over screened elimination weeks, Percent and Rank agree about 0.75 of the time, since Percent preserves magnitude gaps in fan support while Rank retains only order. With Judges' Save, agreement rises (e.g., ≈ 0.89 at $\gamma = 15$) because changes are limited to Bottom-2.

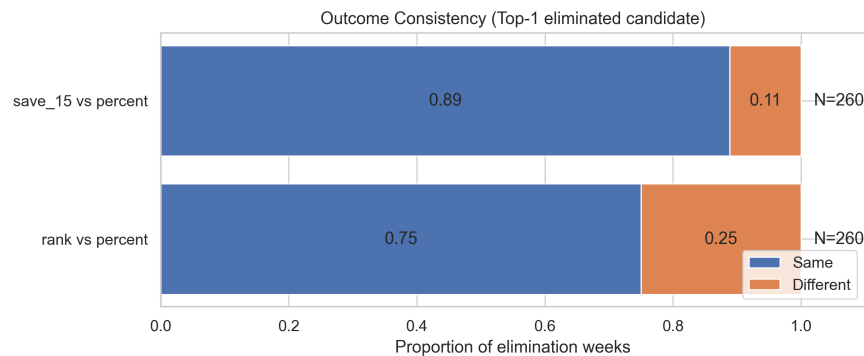


Figure 7: Outcome Consistency

5.3.2 Which rule favors fan votes more? (Influence via absolute correlation)

We measure how each rule transmits fan and judge signals into elimination risk by computing, for each elimination week (s, w) and rule r , within-week Pearson correlations across active contestants^[6]:

$$\rho_{s,w}^{(\text{fan})}(r) = \text{corr}_{i \in \mathcal{A}_{s,w}}(\hat{p}_{\text{elim}}(s, w, i | r), \mu_{s,w,i}), \quad (17)$$

$$\rho_{s,w}^{(\text{judge})}(r) = \text{corr}_{i \in \mathcal{A}_{s,w}}(\hat{p}_{\text{elim}}(s, w, i | r), J_{s,w,i}), \quad (18)$$

where $\mu_{s,w,i}$ is the Q1 mean fan vote share. We compare $|\rho_{s,w}^{(\text{fan})}(r)|$ and $|\rho_{s,w}^{(\text{judge})}(r)|$ across weeks; larger values imply stronger linkage to elimination risk. Empirically, Percent shows the strongest fan influence, Rank attenuates it via rank compression, and Judges' Save shifts influence toward judges as γ increases (Bottom-2 decision).

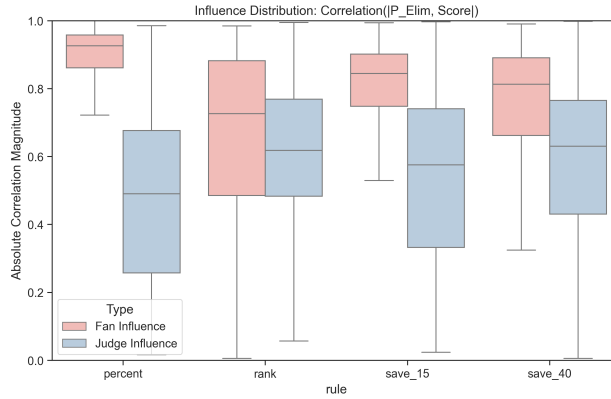


Figure 8: Influence Distribution

5.4 Controversy cases: when judges and fans disagree

5.4.1 Counterfactual survival under alternative rules

We evaluate several well-known controversy targets (fan-favored but judges-disfavored, or vice versa). For each target, we report the simulated *observed-horizon survival probability* under each rule.

Table 4: Survival probability of controversy targets under each rule.

Target	Percent	Rank	Save ($\gamma=15$)	Save ($\gamma=40$)
Jerry Rice (S2)	0.542	0.146	0.226	0.184
Billy Ray Cyrus (S4)	0.076	0.026	0.084	0.028
Bristol Palin (S11)	0.564	0.090	0.152	0.122
Bristol Palin (S15)	0.192	0.030	0.362	0.296
Bobby Bones (S27)	0.672	0.004	0.278	0.238

Two takeaways:

Rules matter. Percent can yield higher survival than Rank for some targets, consistent with stronger fan-signal transmission.

Save is non-monotone. With decisions confined to Bottom-2 and judges' weight increasing in γ , survival can rise or fall with Bottom-2 frequency and judges' score gaps.

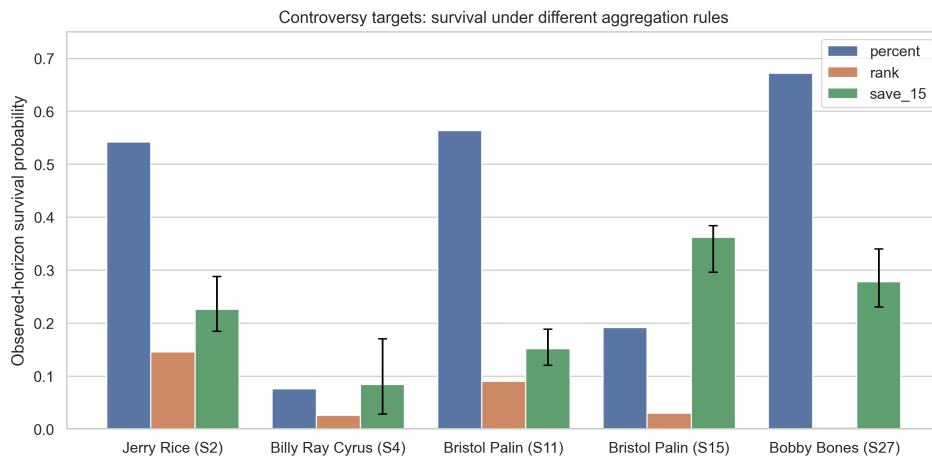


Figure 9: Controversy targets

5.4.2 Mechanism: what changes when Judges' Save is introduced?

Judges' Save first restricts elimination to Bottom-2, so fan support mainly matters for avoiding Bottom-2; within Bottom-2, the decision increasingly follows judges' score gaps as γ increases. Hence, Save- γ moderates fan-driven extremes while retaining fan influence at qualification.

5.5 Recommendation for future seasons

5.6 Recommendation for future seasons

5.6.1 Trade-off summary

The rules form a clear trade-off:

Percent: strongest fan influence (keeps magnitude); **Rank:** more stable outcomes (uses order only); **Judges' Save:** a tunable compromise—fans shape Bottom-2, judges decide within Bottom-2 via γ .

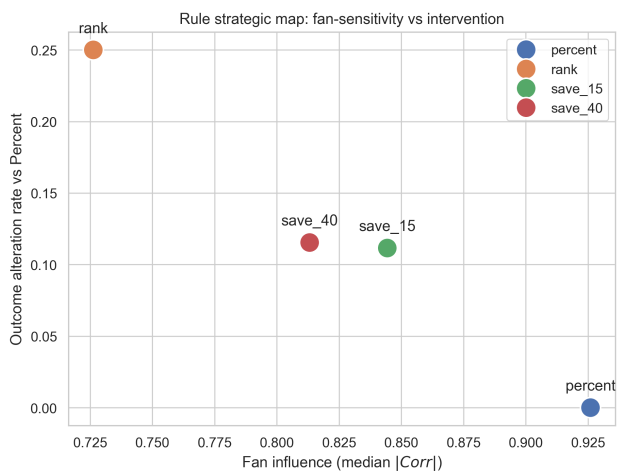


Figure 10: Rule strategic map

5.6.2 Our recommendation

If the goal is an audience-driven show, Percent is most appropriate; if the goal is closer alignment with judges and fewer fan-driven controversies, Rank is safer. We recommend **Judges' Save with moderate γ** (e.g., $\gamma \approx 10\text{--}15$) as a practical compromise: fans still shape Bottom-2 membership, while γ limits extreme outcomes driven by vote magnitude and provides a tunable fairness–entertainment control.

6 Modeling and Solving of Problem 3

6.1 Problem Analysis and Data Structure

We assess whether Pro dancers and celebrity traits affect outcomes, and whether their effects differ between judges' scores and fan votes. The data are hierarchical (contestants nested within weeks; celebrities paired with Pros who recur across seasons), inducing within-Pro correlation. To capture this clustering while retaining interpretability, we use mixed-effects models^[7].

6.2 Variable Construction

To make judges and fan outcomes comparable across weeks (because scoring/voting scales vary by week and season), we standardize within each season-week:

- Judges outcome: J_z = z-score of judges' total score within each (season, week).
- Fan outcome: vote share is transformed by logit and then standardized: V_z .

Celebrity covariates include standardized age (age_z) and industry category (one-hot, with a reference group). We also control week and season fixed effects to absorb systematic time patterns^[8].

6.3 Model Specification

6.3.1 Model A: Judges' channel

$$J_z = \beta_0 + \beta_1 \text{age}_z + \beta_2 \text{industry} + \text{FE}_{\text{week}} + \text{FE}_{\text{season}} + u_{\text{pro}} + u_{\text{celebrity}} + \epsilon. \quad (19)$$

6.3.2 Model B: Fan channel (with performance control)

$$V_z = \alpha_0 + \alpha_1 \text{age}_z + \alpha_2 \text{industry} + \text{FE}_{\text{week}} + \text{FE}_{\text{season}} + \alpha_3 J_z + v_{\text{pro}} + v_{\text{celebrity}} + \eta. \quad (20)$$

The added term $\alpha_3 J_z$ tests whether fans simply follow technical performance, after controlling demographics and time effects.

6.3.3 Model C: Unified difference test (Fans vs Judges)

To formally test whether a factor affects judges and fans in the same way, we stack the two channels into one dataset and estimate:

$$Y = \theta_0 + \theta^\top X + \phi^\top (X \times \mathbb{I}[\text{source} = \text{Judge}]) + \text{FE}_{\text{week}} + \text{FE}_{\text{season}} + u_{\text{pro}} + u_{\text{celebrity}} + \epsilon. \quad (21)$$

Here, θ represents the *fan* baseline effects, and ϕ represents how much the *judge* channel deviates from fans. We treat a difference as meaningful when the interaction term is significant (e.g., $p < 0.05$).

6.4 Results and Interpretation

6.4.1 Pro dancer effects: visible to judges, negligible to fans

The Pro random intercept varies substantially in the judges channel (from -0.251 to $+0.584$, spread ≈ 0.834 standardized), indicating the assigned Pro can systematically shift judges' relative scoring after controlling for week/season and celebrity traits. By contrast, the Pro effect in the fan channel is near zero (-2.16×10^{-4} to $+1.93 \times 10^{-4}$), suggesting fan support is driven primarily by celebrity-level factors rather than the Pro partner.

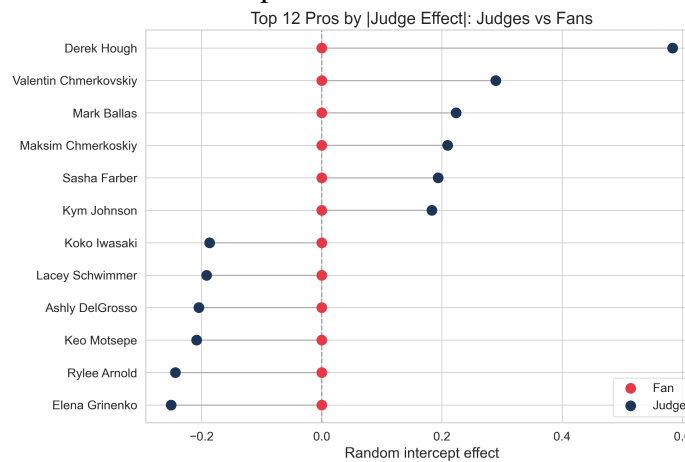


Figure 11: Top 12 Pros by Judge Effect: Judges vs. Fans

6.4.2 Age and industry: fans and judges weight them differently

Age exhibits a strong sign reversal across channels. In the unified model, the fan baseline age effect is positive:

$$\text{Fan age effect} = +0.117 \quad (p < 0.001),$$

while the judge deviation relative to fans is negative:

$$\text{Judge minus fan (age)} = -0.524 \quad (p < 0.001),$$

so the implied total judge-channel age effect is

$$\text{Judge age effect} = +0.117 - 0.524 = -0.406.$$

In words: older contestants receive slightly higher fan support, but judges score them lower after controls.

Industry effects also show systematic divergence. Two examples that are both statistically clear:

- **Other:** fans favor it ($+0.130$, $p < 0.001$), but judges penalize it much more (judge-minus-fan -0.579 , $p < 0.001$), yielding a net judge effect -0.450 .

- **Social:** fans slightly disfavor it (-0.071 , $p = 0.449$), but judges are substantially more positive (judge-minus-fan $+0.350$, $p = 0.008$), yielding a net judge effect $+0.279$.

Overall, age and industry are major sources of “fans vs judges” disagreement.

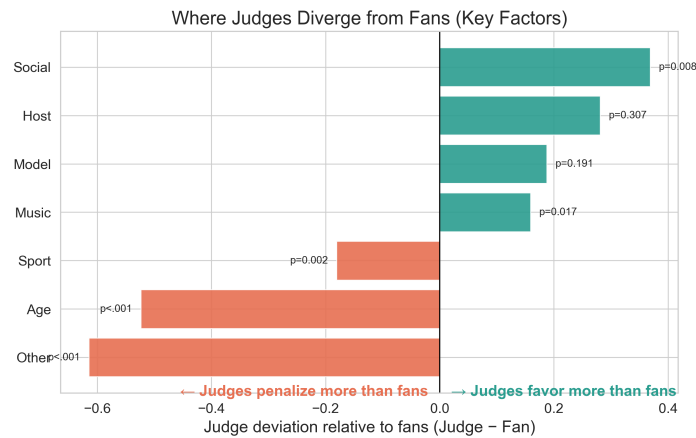


Figure 12: Where Judges Diverge from Fans(Key Factors)

6.4.3 Variance decomposition: who drives outcomes?

Variance decomposition shows that, for fan votes, structured variance is driven almost entirely by celebrities (42.4%), with negligible Pro contribution and 57.6% residual. For judges' scores, celebrities explain 44.2% and Pros 6.9%, with 48.9% residual. Thus, celebrity identity dominates both channels, while Pros matter mainly to judges.

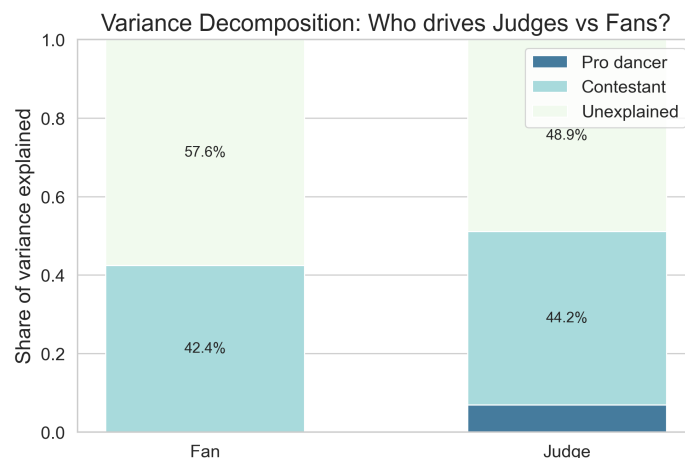


Figure 13: Variance Decomposition: Who Drives Judges vs. Fans?

6.4.4 Do fans reward technical performance the same way as judges?

In the fan model that controls for week/season and demographics, the coefficient of judges' performance is negative:

$$\alpha_3 = -0.149 \quad (p < 10^{-25}).$$

This does *not* mean “dancing well reduces votes” in a causal sense. It indicates that, after accounting for weekly normalization and covariates, fan support contains a strong component not aligned

with judges’ technical scoring (e.g., pre-existing popularity, storyline effects, or audience taste). Therefore, judges and fans do not score contestants in the same mechanism.

7 Modeling and Solving of Problem 4

7.1 Problem Analysis

Question 4 is essentially a *mechanism design* problem: we are not asked to merely replay historical eliminations, but to propose a voting/elimination rule that is (i) fairer under judge–fan disagreement^[9], (ii) less sensitive to extreme fan-vote concentration, and (iii) able to reduce “controversial survivals” while remaining reasonably consistent with the original show logic.

Our technical route is *rule design + uncertainty modeling + Monte Carlo evaluation*. Concretely:

1. **Rule formalization:** implement the proposed mechanism as a computable elimination score;
2. **Simulation-based evaluation:** sample weekly fan-vote shares under uncertainty and run the elimination procedure repeatedly to estimate elimination and champion probabilities;
3. **Diagnostics:** quantify (a) fan-vote extremity mitigation (STF effect), (b) elimination stability and judge–fan conflict, and (c) controversy frequency under explicit thresholds.

7.2 Model Construction

The proposed mechanism contains three components: (i) an **STF** module to suppress extreme fan-vote concentration, (ii) a **judge–fan composite score** after standardization, and (iii) a **Strike & Cap** discipline to constrain persistent low-quality survivals^[10].

7.2.1 STF: Suppressing Extreme Fan-Vote Concentration

Let $X_{s,w,i}$ be the raw fan-vote share of contestant i in season s , week w . We first apply a power transform to compress large shares:

$$X'_{s,w,i} \propto (\tilde{X}_{s,w,i})^\eta, \quad \eta \in (0, 1), \quad (22)$$

so that high shares shrink and low shares are lifted, reducing “vote domination”.

To avoid over-correction, we then mix the transformed distribution with the uniform share $u = 1/|A_{s,w}|$ among active contestants:

$$X^{(\text{stf})} = (1 - \rho)X' + \rho u, \quad \rho \in [0, \rho_{\max}], \quad (23)$$

where ρ increases when the weekly fan-vote distribution is more extreme.

Empirical effect of STF (from weekly aggregates). Across all season-weeks, STF consistently reduces fan-vote extremes. Using *Raw – STF* as “improvement” (positive means extremes are mitigated), we obtain:

- the average reduction in the weekly 90th-percentile of `max_share` is **0.035**;

- the average reduction in weekly mean Gini is **0.052**.

Both improvements are positive across the dataset, indicating STF does not work only on a few outliers; it reshapes the distribution in a stable, directionally consistent way.

7.2.2 Composite Scoring of Judges and Fans

We standardize judges' scores and STF-adjusted fan shares:

$$Z_{s,w,i}^J = \frac{J_{s,w,i} - \bar{J}_{s,w}}{\text{sd}(J_{s,w})}, \quad Z_{s,w,i}^F = \frac{X_{s,w,i}^{(\text{stf})} - \overline{X^{(\text{stf})}_{s,w}}}{\text{sd}(X_{s,w}^{(\text{stf})})}. \quad (24)$$

Then the weekly composite score is

$$S_{s,w,i} = \alpha Z_{s,w,i}^J + (1 - \alpha) Z_{s,w,i}^F, \quad \alpha \in [0, 1]. \quad (25)$$

This keeps fan preference as a central signal, but ensures the two sources are comparable and prevents raw fan-vote scale from overpowering judges by construction.

7.2.3 Fan-Vote Cap with Elimination Discipline (Strike & Cap)

To address “persistently low-quality but high-popularity survivals”, we record a weekly strike when a contestant stays in the bottom tail of judges' ranks:

$$\text{Strike}_i \leftarrow \text{Strike}_i + 1. \quad (26)$$

If strikes exceed a threshold L , we cap the fan component:

$$Z_{s,w,i}^F \leftarrow \min(Z_{s,w,i}^F, C). \quad (27)$$

Finally, if a contestant accumulates too many strikes in the last T weeks (threshold L_{final}), we trigger a hard constraint to prevent “vote-floor”-driven survivals in critical late-stage weeks.

7.2.4 Controversy Diagnostics Metrics

To quantify stability and controversy, we define:

- **Elimination margin** $M_{s,w}$: the boundary gap in elimination probability between the last eliminated and the first safe contestant (larger means more stable);
- **Judge–fan conflict** $C_{s,w}$: average $|Z^J - Z^F|$ among the bottom risk group (larger means stronger disagreement);
- **Controversy week**: a week is flagged if $M_{s,w} < \tau$ and $C_{s,w} > \delta$ with $(\tau, \delta) = (0.2, 1.0)$ in our experiments.

7.3 Model Solution and Results

7.3.1 Overall behavioral summary (one figure, four messages)

Fig. 14 summarizes four key outcomes of NewA:

1. **Controversy frequency:** under $(\tau, \delta) = (0.2, 1.0)$, the controversy-week share is **8.0%** (percent rule), **25.7%** (rank rule), and **10.0%** (NewA). This indicates the rank rule is most prone to unstable, high-disagreement boundary weeks, while NewA stays close to the more stable percent rule and avoids the high controversy rate of rank-based elimination.
2. **Stability and conflict distributions:** NewA shifts the distribution of the boundary margin upward while keeping conflict at a controlled level; in practice this means fewer “coin-flip” eliminations when judges and fans disagree sharply.
3. **STF mitigation (NewA):** STF reduces fan-vote extremes in a consistent direction, with average improvements of **0.035** on $P90(\text{max_share})$ and **0.052** on mean Gini (Raw – STF).

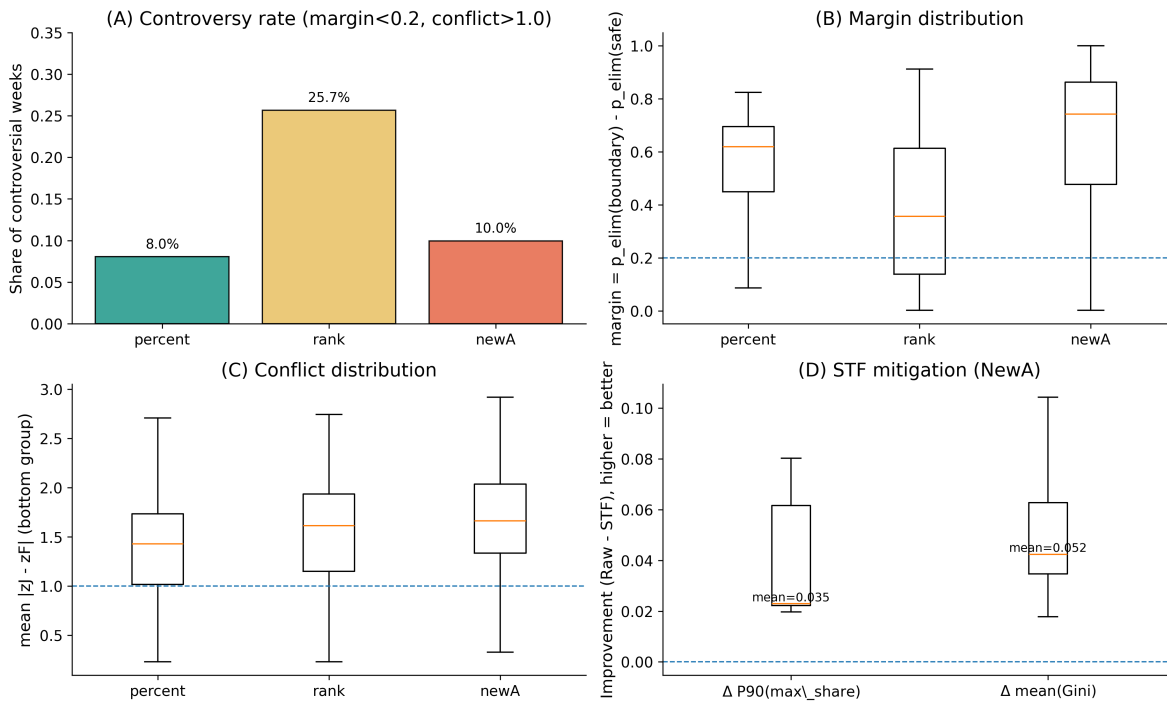


Figure 14: Dashboard summary for Question 4. (A) Controversy-week share under $(\tau, \delta) = (0.2, 1.0)$. (B) Boundary margin distribution. (C) Judge–fan conflict distribution. (D) STF mitigation measured by Raw – STF improvements (positive means weaker fan-vote extremes).

7.3.2 Consistency with original show logic (not a complete rewrite)

Although NewA is designed to intervene on extreme and controversial situations, it does not overturn the weekly trajectory wholesale. When we compare the *weekly highest-risk (argmax) elimination target* under identical inputs, NewA agrees with the percent-based rule in **74.3%** of

weeks and agrees with the rank-based rule in **72.0%** of weeks. This is important: NewA behaves like a “targeted correction” rather than a brand-new game.

7.3.3 Specified controversy cases: NewA corrects unfair survival

We test the cases explicitly highlighted by the problem statement (Bobby Bones, Bristol Palin, Jerry Rice, Billy Ray Cyrus). Using the last $K = 5$ active weeks of each case, NewA assigns substantially higher elimination probability to these “controversial survivors”: **0.37** (Bobby Bones, S27), **0.33** (Bristol Palin, S11), **0.29** (Jerry Rice, S2), and **0.22** (Billy Ray Cyrus, S4). Higher p_{elim} here means the rule is less likely to allow long-term low-quality survival purely due to concentrated fan votes, which matches the design intent of Strike & Cap.

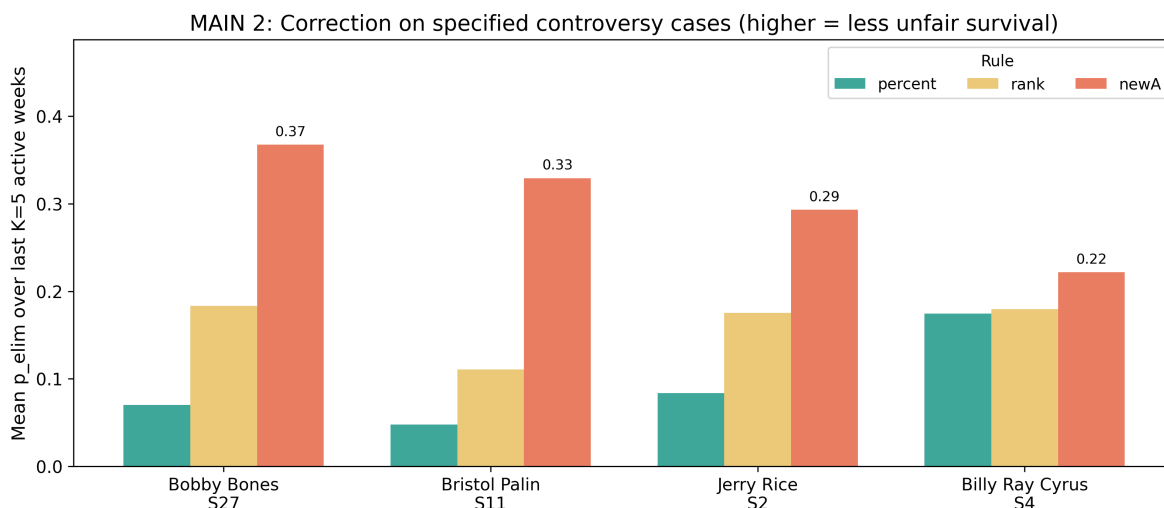


Figure 15: Correction on specified controversy cases (mean elimination probability over last $K = 5$ active weeks). NewA consistently increases elimination risk for the controversial survivors, reducing the chance of “unfair” long-run survival.

7.3.4 Season 27: champion probability becomes less concentrated

Season 27 provides a clear long-horizon comparison. Under NewA (from Monte Carlo champion probability estimation), the top probabilities are distributed across multiple contenders: Milo Manheim (**0.264**), Evanna Lynch (**0.258**), Juan Pablo Di Pace (**0.229**), and Alexis Ren (**0.212**). Meanwhile, the overall concentration measured by $HHI = \sum_i p_i^2$ drops to **0.234** under NewA, compared with **0.708** (percent) and **0.858** (rank). A smaller HHI means the outcome is less “locked in” by one contestant, which is precisely what STF aims to prevent when fan-vote shares become extreme.

7.3.5 Conclusion for Question 4

Overall, NewA improves the mechanism along three practical dimensions:

1. **Fairness:** controversial survivors receive higher elimination risk in late stages (e.g., 0.37/0.33/0.29/0.22 on the specified cases), reducing “vote-floor” survivals under poor judge performance.

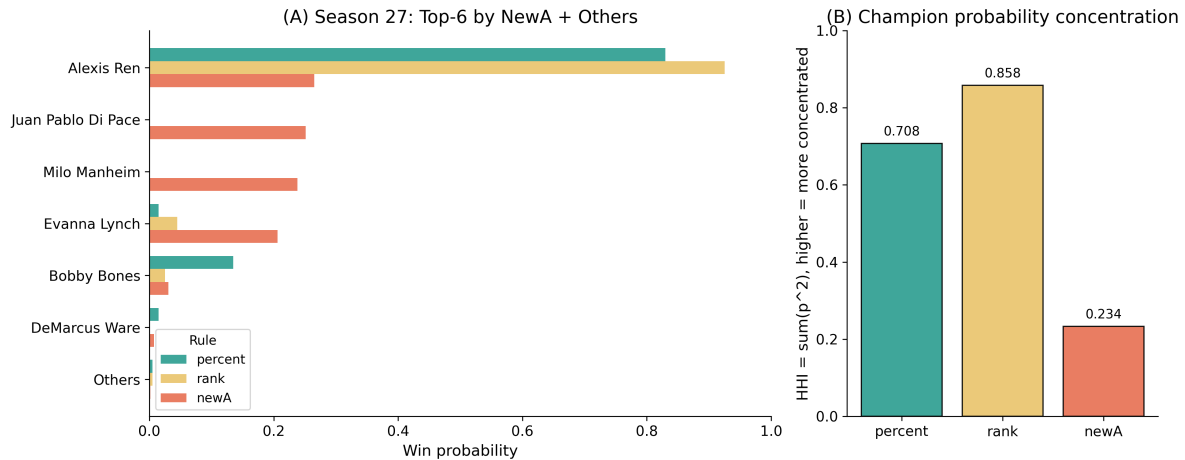


Figure 16: Season 27 champion probability comparison. (A) Top-6 by NewA plus an “Others” bucket. (B) Probability concentration measured by $HHI = \sum p^2$ (higher means more concentrated). NewA produces a markedly less concentrated forecast.

2. **Representativeness with restraint:** fan preference remains decisive, but STF reduces extreme concentration by a stable margin (0.035 on $P90(\text{max_share})$ and 0.052 on mean Gini).

3. **Stability and controversy control:** NewA avoids the high controversy rate of rank-based elimination (25.7%) and stays close to the stable percent rule (8.0%), yielding 10.0% under the same thresholds.

These results support NewA as an implementable and quantitatively justified revision for the show’s elimination mechanism.

8 Sensitivity Analysis (Fan prediction model)

We test robustness by perturbing key Q1 hyperparameters and re-running weekly inference: the margin Δ (percent rule), concentration γ (peakedness of $\mathbf{v}_t$), and prior weight W_{prev} (remaining weight assigned to the season baseline with a fixed uniform anchor). We report Bottom-1/Bottom-2 consistency, stability $\mathbb{E}\|\mathbf{v}_t - \mathbf{v}_{t-1}\|_1$, concentration (HHI), mean 95% CI width, and slack usage (Fig. S1).

Across all settings, Bottom-2 hit rate remains 1.0, Bottom-1 accuracy stays near 0.99–1.00, solver success is 100%, and slack is unused, indicating low sensitivity of the core conclusions. The main variation is in stability and HHI: larger Δ , γ , or W_{prev} yields smoother and/or more concentrated vote-share trajectories without materially changing elimination consistency.

Fig. S1 caption (short): **Figure S1.** Q1 sensitivity dashboard: metrics versus Δ , γ , and W_{prev} . Consistency is stable, while stability and HHI vary as expected.

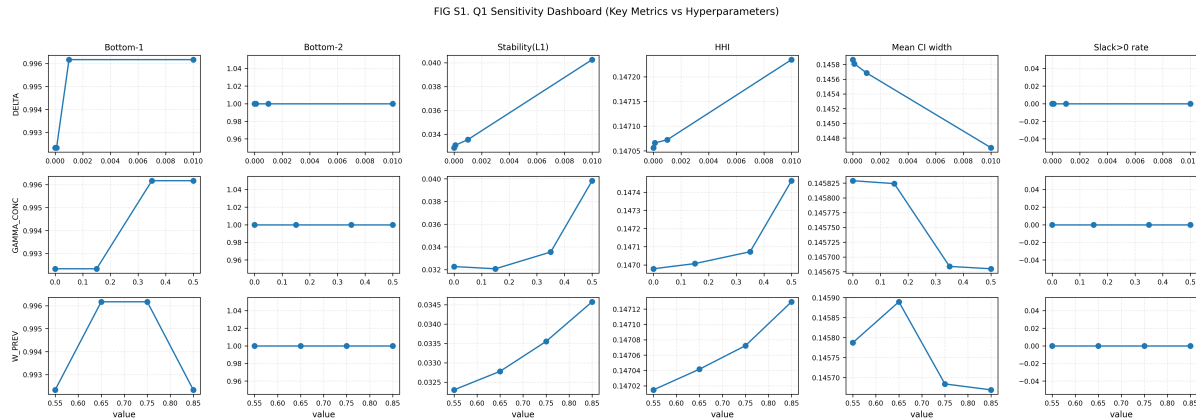


Figure 17: Season 27 Champion Probability Comparison

9 Model Evaluation

9.1 Strengths

1. **Strong scenario adaptability:** Tailored to DWTS's rule characteristics, with key parameters having clear practical meanings, aligning with the show's operation logic.
2. **Comprehensive uncertainty handling:** Integrates Beta distribution fitting, constraint-aware sampling, and weighted percentile estimation to quantify reliability of inferred vote shares and simulation outcomes, avoiding deterministic biases.
3. **Robust hierarchical data modeling:** Mixed-effects model captures clustered data features (e.g., pro-contestant pairing), and the unified interaction model enables rigorous testing of judges-fans preference differences, enhancing result credibility.

9.2 Weaknesses and Improvements

The model simplifies complex real-world factors, including dynamic drivers of fan preference (e.g., social media sentiment), multi-dimensional judges' decision-making criteria, and unmodeled contestant characteristics (e.g., pre-show popularity), leading to potential residual uncertainty and limited generalization. Integrate external data (e.g., social media trends, historical judges' save records) to refine dynamic fan preference modeling and judges' decision mechanisms, expand characteristic variable dimensions, and validate the framework across similar reality TV competitions to enhance accuracy and generalizability.

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A Letter to the Production Team of *Dancing with the Stars*

Dear Esteemed Production Team of *Dancing with the Stars*:

We are researchers focused on reality-show mechanism design. To balance professional authority, audience participation, and fairness in DWTS, we share a refined voting-integration proposal on behalf of our team.

We have analyzed past seasons' voting rules, and our framework balances entertainment value with result credibility:

Recommended Voting Integration Plan

Our recommended voting integration plan includes three key adjustments: for regular episodes, we suggest a 60% weight for judges' scores paired with 40% for audience votes, shifting to a 50/50 split in the finale to amplify audience input for the championship; to prevent extreme voting outcomes, we propose capping individual audience vote shares at 25% and reducing the audience weight by 10% for contestants with judges' scores in the bottom 20%.

Timeline for Future Seasons

S35: Pilot 60/40 weight + 25% vote cap; adjust parameters via real-time feedback.

S36: Add low-score weight adjustment; track professional–audience balance via post-episode data.

We trust this proposal offers a clear, practical approach to blending judges' scores and audience votes effectively. Implementing these adjustments could strike the perfect balance between professional credibility and viewer engagement—key to keeping *Dancing with the Stars* the vibrant, beloved competition it has always been.

Looking forward to seeing these ideas bring even more excitement and fairness to future seasons of the show.

Sincerely,
Team #2622052

Report on Use of AI

1. OpenAI ChatGPT (Jan 30, 2025 version, ChatGPT5.2-Thinking)

Query: What is maximum entropy? Please give me a brief explanation.

Output: Maximum entropy is the idea that, when you have limited information about something, the best “default” probability model is the one that adds the fewest extra assumptions—i.e., the one with the highest entropy among all models that satisfy what you do know.

- Entropy (in information theory) measures how spread out / uncertain a probability distribution is.
- So the maximum entropy distribution is the most “uninformative” or “least committed” distribution that still matches your constraints (like a known mean, total probability = 1, known averages, etc.).

Why it’s useful: it gives a principled way to choose a distribution from many possibilities without sneaking in bias.

Tiny example: If all you know is that a variable can take one of n outcomes and nothing else, the max-entropy choice is uniform: each outcome has probability $1/n$.

2. OpenAI ChatGPT (Jan 31, 2025 version, ChatGPT5.2-Thinking)

Query: What is Monte Carlo simulation? Please provide a brief explanation.

Output: Monte Carlo simulation is a method that uses random sampling and repeated trials to estimate outcomes in problems with uncertainty, especially when an exact analytical solution is hard or impossible.

- Model uncertain inputs (e.g., cost, demand, time) with probability distributions.
- Randomly sample those inputs many times (often thousands or millions).
- Compute the outcome each time.
- Aggregate the results to estimate the expected value, variability, and probabilities.

In one sentence: it’s “simulate randomness many times to approximate the real-world distribution of outcomes.”

3. OpenAI ChatGPT (Feb 1, 2025 version, ChatGPT5.2-Thinking)

Query: What is the core idea of the mixed-effects model?

Output: A mixed-effects model (a.k.a. multilevel/hierarchical model) combines:

- **Fixed effects:** effects assumed to be the same across the whole population (e.g., the average impact of a treatment).
- **Random effects:** effects that vary by group/subject and are modeled as drawn from a probability distribution (e.g., each hospital or each person has its own baseline level).

Core idea: model both the overall trend and group-specific deviations at the same time, so the model can handle correlated/clustered data (e.g., repeated measures on the same person, students within schools, patients within hospitals).

A common simple form:

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + u_j + \varepsilon_{ij}.$$

- β_0, β_1 : fixed effects (population-level)
- u_j : random effect for group j (group-level shift)
- ε_{ij} : residual noise

Why it's useful: it shares information across groups ("partial pooling"), improving estimates—especially when some groups have little data.